

Executive Decision Report

Retail Sales & Business Intelligence Analysis

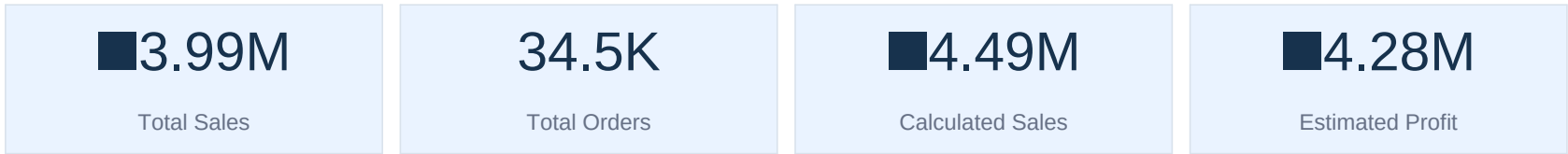
Interactive Power BI Dashboard | Capstone Presentation

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1. Executive Summary

What management should know

The dashboard indicates a strong retail operation with revenue concentrated in Electronics, a leading South region, broadly balanced customer-gender contribution, and a measurable return rate. The main management opportunity is to protect high-value categories while improving return prevention and monthly performance control.



- Electronics is the leading visible category by sales.
- South is the strongest visible region.
- Returned orders are approximately 5.52% of total orders.
- Female and male customers contribute almost equally.
- Management should focus on category growth, return reduction, and monthly monitoring.

2. Business Objective & Methodology

Objective: convert cleaned transaction data into executive-level decisions related to revenue, orders, category performance, regional performance, customer behavior, payment methods, and operational risk.

Step	Activity	Purpose
1	Data preparation	Use the cleaned dataset from earlier internship tasks.
2	KPI calculation	Review sales, orders, calculated sales, and estimated profit.
3	Comparative analysis	Compare category, region, gender, payment, and return patterns.
4	Trend analysis	Review monthly sales movement using order-date month.
5	Relationship analysis	Use scatter analysis for price, quantity, and calculated sales.
6	Decision framing	Convert patterns into recommendations and an implementation plan.

Limitation: ROI values in this report are planning scenarios, not audited financial forecasts.

3. KPI Snapshot

KPI	Dashboard Value	Executive Interpretation
Total Sales	■3.99M	Total amount across transactions.
Total Orders	34.5K	Count of order IDs.
Calculated Sales	■4.49M	Calculated sales measure used in the dashboard.
Estimated Profit	■4.28M	Estimated profit measure used for directional analysis.

Control recommendation: Clearly define whether each sales measure is gross, net, after discount, or after returns.

4. Key Revenue Drivers

Area	Observed signal	Management action
Electronics	Highest category contribution	Protect stock, monitor pricing, and use complementary-product bundles.
Home / Sports / Fashion	Secondary contributors	Use targeted campaigns and compare margin and return behavior.
South	Highest regional sales	Study assortment, customer mix, and payment preferences.
North / East / West	Strong regional contribution	Replicate successful practices where appropriate.
Central	Lowest shown region	Investigate demand, assortment, delivery, and marketing reach.

5. Customer & Payment Insights

Dimension	Dashboard interpretation	Decision implication
Customer gender	Female and male contribution is close; Other is smaller.	Use behavior and category preference rather than gender alone.
Credit Card	Largest payment-method sales contribution.	Maintain reliable authorization and checkout support.
Debit Card / UPI	Important secondary channels.	Monitor payment success and checkout drop-offs.
COD	Meaningful contribution with operational friction.	Track cancellation, delivery failure, and return cost.
Wallet / PayPal	Smaller displayed contribution.	Use incentives only after measuring incremental sales.

6. Risk Areas & Operational Signals

Risk	Evidence	Business impact	Control
Returns	Approx. 5.52% returned orders	Refund and reverse-logistics cost	Track by category, region, product, and delivery time.
Monthly fluctuation	Sales rises and falls across months	Unstable planning	Use monthly alerts and investigate large changes.
Metric definition	Total and calculated sales differ	Reporting confusion	Document and reconcile KPI formulas.
Regional imbalance	Central trails stronger regions	Uneven market performance	Review demand, delivery, and marketing.
Payment dependence	Credit Card leads mix	Channel concentration	Monitor failures and keep alternatives.

7. Three Strategic Recommendations

Recommendation 1: Protect and expand high-performing categories

- Prioritize Electronics inventory and margin monitoring.
- Create bundles and cross-sell offers.
- Review price, discount, quantity, and return patterns weekly.

Recommendation 2: Reduce avoidable returns

- Build return analysis by category, region, product, and delivery time.
- Investigate products with unusually high returns.
- Improve product descriptions and delivery communication.

Recommendation 3: Build monthly performance controls

- Set monthly sales and order targets.
- Add alerts for large month-over-month changes.
- Review payment-method performance and checkout failures.

8. Illustrative ROI Projection

These assumptions help management evaluate the ideas. They are not measured dataset results. Replace them with validated baseline costs and benefits before approving a budget.

Initiative	Scenario assumption	Expected value
Return reduction	Reduce returned-order rate by 1 percentage point	Lower reverse-logistics and refund leakage.
Category growth	Increase high-performing category sales by 3%	Incremental revenue from focused campaigns and availability.
Dashboard controls	Reduce reporting/review delay by 20%	Faster response to weak months and operational issues.

ROI formula: $ROI \% = (\text{Incremental benefit} - \text{implementation cost}) \div \text{implementation cost} \times 100$

9. Implementation Timeline

Time	Phase	Key deliverable
Week 1	Baseline & definitions	Confirm KPI dictionary and return-rate baseline.
Weeks 2–3	Category and return analysis	Identify high-value categories and high-return products.
Weeks 4–5	Pilot actions	Run one category campaign and one return-reduction pilot.
Weeks 6–8	Dashboard controls	Add monthly alerts, regional targets, and payment monitoring.
Week 9+	Review & scale	Compare with baseline and scale successful actions.

Governance: Assign a business owner, data analyst, and operations representative. Review weekly and hold a formal monthly decision meeting.

10. Executive Dashboard Design

Dashboard area	Required content
Top row	Total Sales Total Orders Calculated Sales Estimated Profit
Middle row	Sales by Category Sales by Region Sales by Customer Gender Monthly Sales Trend
Bottom row	Returned vs Non-Returned Orders Sales by Payment Method Sales by Region and Gender
Interactive controls	Category filter Region filter Date/month filter Drill-through to detail

Current status: The Power BI dashboard includes KPI cards, category and region charts, customer-gender donut, monthly trend, return analysis, payment-method chart, scatter chart, and category/region filters.

11. Final Decision Summary

Management should use the dashboard as a recurring decision tool rather than a one-time report.

- Protect Electronics and other high-contribution categories through inventory, pricing, and cross-selling controls.
- Treat returns as a measurable operational cost and analyze them by category, region, and delivery experience.
- Use monthly trend monitoring to identify performance changes early.
- Keep payment options diversified while monitoring the leading Credit Card channel.
- Standardize KPI definitions so Total Sales, Calculated Sales, and Estimated Profit are clearly understood.
- Validate financial benefits against a baseline before claiming ROI.

Conclusion: The dashboard provides a strong foundation for data-driven retail decisions. The next step is to operationalize the insights through owners, targets, recurring reviews, and measured pilots.

Appendix: Evidence Used

Evidence source	Used for
Dashboard KPI cards	Total Sales, Total Orders, Calculated Sales, Estimated Profit
Category chart	Electronics is visibly the leading category
Region chart	South is visibly the strongest region
Customer chart	Female and male contribution is broadly balanced
Monthly trend	Visible month-to-month fluctuations
Return chart	Returned orders approximately 5.52%
Payment chart	Credit Card is the largest displayed channel
Scatter chart	Price and calculated sales relationship by category

Prepared for internship mentor evaluation. Replace scenario-based ROI assumptions with validated financial inputs before external business use.